Linear and Nonlinear Kalman Filtering: theory and applications

SCUDO/DAUSY/DRIM Ph.D. course A.Y. 2026

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Outline

1. General info about the course and Introduction
2. Preliminaries
3. Linear Theory
   - State Space models
   - The Kalman Filter(s) and Predictor
   - Optimality and Asymptotic Theory
4. Nonlinear Theory
   - Extended Kalman Filter
   - Unscented Kalman Filter
   - Particle Filters
5. Adds-on
   - Filter Tuning
   - Kalman Smoother
   - Control perspective: LQG

1. General info about the course and Introduction

Schedule

Date	Time	Tentative Topic
Mon. 27/04/2026	h10.30-12:30 (2h)	Intro e Preliminaries
Mon. 04/05/2026	h09.30-12:30 (3h)	Preliminaries/Linear Theory
Mon. 11/05/2026	h09.30-12:30 (3h)	Linear Theory
Mon. 18/05/2026	h09.30-13:30 (4h)	Linear Theory + Essay
Mon. 25/05/2026	h09.30-13:30 (4h)	Linear/Nonlinear Theory + Essay
Mon. 01/06/2026	h09.30-13:30 (4h)	Nonlinear Theory/Adds-on + Essay

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Fri. 22/05/2026	h09.30-13:30 (4h)	Linear/Nonlinear Theory + Essay
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Exam

Practical coding essay + report:

- on a selected application: pick a topic possibly related to your need/research.
- partially developed during class.
- programming language: python – either using jupyter notebooks (recommended) or script based. Make sure all necessary dependencies to run are available.
  For other programming language ask first!!
- no use of predefined libraries with KF implementations.
- you can work in a group if preferred.
- **objective**: explore (possibly all) the topics covered during the course, from linear to non-linear KF. You might need to start from modeling assumptions, linearize if necessary, discuss noise statistics and then implement the filter. Try to explore limits and nuances as if it were a small research project.
- present results in a pdf report[1]

[1] For well organized runnable Jupyter notebooks this is optional. If, upon agreement, another language is used this is mandatory.

References and additional material

IT **Picci, G. (2007). Filtraggio statistico (Wiener, Levinson, Kalman) e applicazioni. Libreria Progetto.**

EN Sanso F., Pileggi, C. and Biagi, L. (2024). A tutorial on Kalman filtering. The linear, extended and unscented filters. (pdf available as resource in the material folder)

EN Anderson, B. D., and Moore, J. B. (2005). Optimal filtering. Courier Corporation.

G-drive:
`https://drive.google.com/drive/folders/1BbuSTiO4wym0qOAqeLUkiVum_14kRRzD`

github: for complete slides, code and additional resources
`https://github.com/MarcoTodescato/linear_and_nonlinear_kalman_filtering`

For Q/A and discussion we can set up a Discord channel

Goal

How to formulate estimation problems encountered in many engineering fields and applications as instances of a general dynamical system state estimation problem and how to derive the mathematical solution of the latter problem (possibly depending on additional assumption on the functional family of the considered dynamical system).

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Essentially, this will lead to the well-known (linear) Kalman filter and its (nonlinear) extensions.

A bit of history: the theories of statistical filtering

Wiener-Kolmogorov (1940)

- $I = (-\infty, t]$, infinite time support
- $\{\mathbf{x}(t)\}$, $\{\mathbf{y}(t)\}$ are weakly jointly stationary

The impulse response of the optimal filter is time-invariant hence it can be analytically computed once and for all times, knowing the processes statistics.

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- no more infinite (lower unbounded) time support

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Kalman (1960)

- signals can be modeled as stochastic dynamical systems of finite size

Looks for recursive finite-memory schemes that resembles that same dynamical system structure of finite size.

A bit of history: a change of perspective

Revolutionary apporach introduced by R.E. Kalman in 1960 to the problem of statistical filtering.

Change of perspective:

- **Wiener-Kolmogorov**: explicit computation of the filter **transfer function**, practically useful only for signals characterized by a **rational** spectral density
  - implicit computations possible only for rational signals
  - actual physical implementation of the filter as a set of difference (recursive) equations only for rational representations

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- **Kalman**: hypothesis that the signals can be described as (linear) **dynamical systems** of finite dimension
  - no transfer functions involved
  - natural formulation of the filter as a fixed algorithmic structure essentially consisting of a discrete-time (linear) dynamical system of finite memory, driven by measurements
  - implementation always possible as a recursive algorithm

Prediction vs Filtering vs Smoothing

Assume $\{\mathbf{x}(t)\}$ and $\{\mathbf{y}(t)\}$ are (second order[2]) processes of known dimensions. Denote with $\hat{\mathbf{x}}(t|I)$ the estimator of $\mathbf{x}(t)$ given $\{\mathbf{y}(t)\}_{t \in I}$, I being the time support.

It is possible to identify 3 classes of **estimation problems**.

[2] First and second moment statistics exist and are finite.

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Find $\hat{\mathbf{x}}(t+h|I)$ being $I=[t_0,t]$ and $h>0$, that is the estimator h-step ahead in the future from measurements up to time t.

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Smoothing

Find $\hat{\mathbf{x}}(t|I)$ being $I = [t_0, t_1]$ with $t < t_1$, that is the estimator at time t from all available measurements up to t_1 in the future.

[2] First and second moment statistics exist and are finite.

2. Preliminaries: before diving in... some definitions and preliminary results

Random signals and stochastic processes

Random signal

A random signal can defined as the tuple (T, V, S) where

- T is the set of time. We consider discrete-time signals, $T = \mathbb{Z}$
- V is the "alphabet" of signal values, $V = \mathbb{R}$ or $V = \mathbb{R}^n$
- S is the sample space. Every $s \in S$ is a function $s : t \mapsto s(t)$ where $t \in T$ and $s(t) \in V$ also called *trajectory* of the signal. S specifies the a-priori admissible trajectories for the signal.

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Stochastic process

Mathematical formulation/description used to formally model the aleatoric nature of a random signal. Consisting of the tuple $(S, \mathcal{S}, P)$ where:

- S sample space. Set of all possible trajectories
- $\mathcal{S}$ set of all logical meaningful propositions or events regarding the signal
- P probability measure on $\mathcal{S}$, i.e. $P : \mathcal{S} \mapsto [0, 1]$ mapping every $A \in \mathcal{S}$ to $P(A) \in [0, 1]$, the a-priori "confidence" A actually happening

Stochastic process: weather example

Suppose we observe weather over 2 consecutive days where weather can be:

- σ: sunny
- ρ: rainy

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- $\mathcal{S}$ could be $\{E_1, E_2\}$ where
  - E_1: *it rains at least one day* $:= \{(\sigma,\rho),(\rho,\sigma),(\rho,\rho)\}$
  - E_2: *the first sunny day* $:= \{(\sigma,\sigma),(\sigma,\rho)\}$

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  - E_2: *the first sunny day* $:= \{(\sigma, \sigma), (\sigma, \rho)\}$
- P: assume weather is independent across days with $P[\sigma] = 0.7$ and $P[\rho] = 0.3$. Then
  - $P[E_1] = (0.7 * 0.3) + (0.3 * 0.7) + (0.3 * 0.3) = 0.51$
  - $P[E_2] = 0.7^2 + (0.7 * 0.3) = 0.7$

Stochastic process: robot navigation example

Assume there is a small robot moving on a 1D line. At each time it can either

- move to the right: $+1$
- stay still: 0

We observe trajectories of 2-steps (x_0, x_1, x_2) starting from $x_0 = 0$.

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- P: assume independent movements $P[0] = 0.7$ and $P[+1] = 0.3$. Then
  - $P[E_1] = (0.3 * 0.7) + (0.3 * 0.3) + = 0.3$
  - $P[E_2] = 0.7^2 = 0.49$

Obviously this can easily be made more complicated to describe more meaningful events.

A more general formulation

Stochastic process (usual def in prob. theory)

A collection of random variables, $\{y(t)\}$, indexed by a temporal index $t \in T$, all defined on the same (abstract) probability space $(\Omega, \mathcal{A}, P)$ where:

- Ω event set of "state of nature" ω
- $\mathcal{A}$ σ-algebra of subsets of Ω
- P probability measure on $\mathcal{A}$

Each variable is a ($\mathcal{A}$-measurable) function $y(t) : t \mapsto y(t, \omega)$. The *trajectories* are the successive determinations $\{y(\cdot, \tilde{\omega})\}$ of $y(t)$ corresponding to a particular event $\tilde{\omega} \in \Omega$

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Our definition $(S, \mathcal{S}, P)$ is a "concrete path-based" version of the abstract $(\Omega, \mathcal{A}, P)$. If we define a variable $y(t) \in (S, \mathcal{S}, P)$ as

$$y(t, s) := s(t)$$

where $y(t)$ associates at every trajectory $s \in S$ its value at t, then it is clear that, for varying t, the successive determinations of the variables $\{y(t)\}$ corresponding to the event $\bar{s} \in \mathcal{S}$ describe indeed the trajectory $\{\bar{s}(t)\}$ of the process.

Let's clarify...

The probabilistic formulation and formalization introduced is *just* a convenient choice to describe (or model) the unpredictable nature of a signal.

It is worth remembering that this is one possible choice that eventually allows one to resort to tools from probability theory to describe, handle and develop useful tools for filtering and identification.

For this reason, it is important to understand that, regardless the real nature of the underline signal, the same formulation (as *stochastic process*) can indeed be used to describe every signal (aleatoric or not).

On the other hand, it might be worth remembering that this choice is dictated by the "simplicity", compactness and generality to describe events that could indeed also be described using a "deterministic" formulation.

The Problem of *Estimation*: Estimate vs Measure

Statistical problem of estimation seen as formalization of the mathematical problem of measuring where the goal is that of getting information about a desired quantity x, non directly observable, based on observable data, y, whose values are produced by the measurement "instrument".

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Ideally, knowing the “physical coupling”, i.e., relation, between x and y should allow to retrieve the value of x based on observable values of y. Practically, this relation is either not perfectly known or influenced by *environmental conditions* which introduce *uncertainty* in the measurement process.

$$y = h(x) \Longleftrightarrow y = h(x) + \epsilon$$

As a consequence, the objective can be understood (or interpreted) as that of *estimating* the value of x based on the observed values of y.

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Following this "uncertainty-based" interpretation:

- it is not always straightforward how to generally get x from y
- being impractical (meaningless) to reconstruct x with certainty, how to *best* approximate it?

The Bayesian approach

x assumed to be a random quantity characterized by some a-priori information

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Notation: random variables vs sample values

It is important to understand the difference between a random variable, for which bold letters $\mathbf{x}$ will be used vs the corresponding sample values, i.e., the actual value taken by the random variable, for which normal case letters x will be used.

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Bayesian inference

Assume there exist a random variable/vector $\mathbf{x}$ described by the probability density $p_{\mathbf{x}}(x)$ and the random variable/vector $\mathbf{y}$ of measurements described by the conditional density $f(y|x) = f(y|\mathbf{x} = x) = \frac{P(y \leq \mathbf{y} \leq y + dy | \mathbf{x} = x)}{dy}$.

The problem of Bayesian inference/estimation is that of reconstructing $\mathbf{x}$ based on $p_{\mathbf{x}}$, $f(y|x)$ and the observation of $\mathbf{y} = y$. This can be understood as to:

- compute the new distribution of $\mathbf{x}$ determined by the observation $\mathbf{y} = y$, or
- reconstruct $x = \mathbf{x}(\omega)$ determined by the "measurement" process

Bayesian inference: complete reconstruction

Assuming $p_{\mathbf{x}}(x)$ and $f(y|x)$ completely known, the problem is that of computing

$$f(x|y) = \frac{f(y|x)p_{\mathbf{x}}(x)}{\int f(y|x)p_{\mathbf{x}}(x)dx} = \frac{p_{\mathbf{yx}}(y,x)}{p_{\mathbf{y}}(y)}$$

i.e., the conditional *a-posteriori* density of $\mathbf{x}$ given $\mathbf{y} = y$, which is the most exhaustive and complete description possible.

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In practice, this is rarely the case and/or even necessary because rather than an exhaustive description, a more handy point estimate of $\mathbf{x}$, telling us what is the unobservable sample value $\mathbf{x} = x$ that gave rise to the observation $\mathbf{y} = y$, is more useful.

Bayesian inference: point estimate

Assume the relation between $\mathbf{x}$ and $\mathbf{y}$ could analytically be described as

$$\mathbf{y} = h(\mathbf{x}, \mathbf{w})$$

being $\mathbf{w}$ the random vector capturing the variable interations with the environment, i.e., the "measurement noise".

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During the measurement process the particular experimental conditions ω give raise to a specific determination $\mathbf{w}(\omega) = w$ which, in turn, give raise to an observed value $\mathbf{y} = y$ linked to $\mathbf{x} = x$ by the relation

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Then, the problem is that of reconstructing the value x of $\mathbf{x}$ that "caused" $\mathbf{y} = y$ during the measurement process. Since estimating/knowing w is intrinsically impossible, the problem is understood as that of "best" approximating x.

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It is then necessary to define a "reasonable" class of approximation criteria.

Bayesian inference: conditional/posterior expected risk

Cost/Loss Function

It is called "loss/cost" a scalar function $c : \mathbb{R}^n \mapsto \mathbb{R}$ of $\xi = z - x$, $z \in \mathbb{R}^n$, s.t. $c \geq 0$, $c'' > 0$ (strictly convex) and $c(0) = 0$. It is said symmetric if $c(-\xi) = c(\xi)$.

E.g.: $c(z - x) = \|z - x\|_Q^2$ where $Q > 0$ symmetric and $\|x\|_Q^2 = x^T Qx$.

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z plays the role of approximation of x. The problem could then be that of finding z that minimizes c for every value of the unknown x. In practice it is "smarter" to consider all the available probabilistic information, e.g., given by $\mathbf{y} = y$.

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Conditional Expected Risk and Point Bayesian Estimate

Consider the *Conditional Expected Risk*

$$R(z, y) := \mathbb{E}[c(z - \mathbf{x}) | \mathbf{y} = y] = \int_{\mathbb{R}^n} c(z - x) f(x|y) dx$$

Then, the *Point Bayesian Estimate* is defined as

$$\hat{x} = \text{Arg}\ \min_z R(z, y)$$

Bayesian inference: Conditional Expectation

Theorem: Conditional Expectation

If the loss c is symmetric and the conditional density $f(\cdot|y)$ is symmetric around its mean $\mu(y) = \mathbb{E}[\mathbf{x}|\mathbf{y} = y] = \int x f(x|y) dx$, then $\hat{x}$ does not depend on c and it is equal to the conditional mean of $\mathbf{x}$ given $\mathbf{y} = y$, i.e., $\mathbb{E}[\mathbf{x}|\mathbf{y} = y]$.

Moreover, if the cost c is of quadratic form, i.e., $\| \cdot \|_Q^2$, then $\hat{x} = \mathbb{E}[\mathbf{x}|\mathbf{y} = y]$ for all types of conditional density $f(x|y)$ (assuming the conditional mean exists).

More generally, among all possible measurable functions $g : \mathbb{R}^m \mapsto \mathbb{R}^n$, for every symmetric $Q > 0 \in \mathbb{R}^{n \times n}$,

$$\mathbb{E}[\mathbf{x}|\mathbf{y}] = \text{Arg min}_{g(\cdot)} \mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2]$$

[3] Estimator: function (algorithm) of the observed data mapping them to point estimates.

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$$\mathbb{E}[\mathbf{x}|\mathbf{y}] = \text{Arg}\min_{g(\cdot)} \mathbb{E}[\|\mathbf{x} - g(\mathbf{y})\|_Q^2]$$

Interpreted as a function of y, for symmetric c and f, the conditional mean $\mathbb{E}[\mathbf{x}|\mathbf{y}] : y \mapsto \mathbb{E}[\mathbf{x}|\mathbf{y} = y]$ is the Bayesian estimator[3] $\hat{x}$ of x that minimizes the conditional expected risk $R(z, y)$. This is also true for any f if c is quadratic. In this case it is referred to as the *minimum squared Bayesian estimator*.

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Theorem: Conditional Expectation (proof of last part)

For the law of total expectation, we can write

$$
\begin{aligned}
\mathbb{E}[\|\mathbf{x}-g(\mathbf{y})\|_Q^2] &= \int_{\mathbb{R}} p_{\mathbf{y}}(y)\mathbb{E}[\|\mathbf{x}-g(\mathbf{y})\|_Q^2|\mathbf{y}=y]dy \\
&= \mathbb{E}\left[\mathbb{E}[\|\mathbf{x}-g(\mathbf{y})\|_Q^2|\mathbf{y}]\right] \\
&= \mathbb{E}\left[\mathbb{E}[\|\mathbf{x}-\mathbb{E}[\mathbf{x}|\mathbf{y}]+\mathbb{E}[\mathbf{x}|\mathbf{y}]-g(\mathbf{y})\|_Q^2|\mathbf{y}]\right] \\
&= \mathbb{E}\Big[\mathbb{E}[[\mathbf{x}-\mathbb{E}[\mathbf{x}|\mathbf{y}]]^T Q[\mathbf{x}-\mathbb{E}[\mathbf{x}|\mathbf{y}]]|\mathbf{y}] \\
&\quad + 2\mathbb{E}[[\mathbf{x}-\mathbb{E}[\mathbf{x}|\mathbf{y}]]^T Q[\mathbb{E}[\mathbf{x}|\mathbf{y}]-g(\mathbf{y})]|\mathbf{y}] \\
&\quad + \mathbb{E}[[\mathbb{E}[\mathbf{x}|\mathbf{y}]-g(\mathbf{y})]^T Q[\mathbb{E}[\mathbf{x}|\mathbf{y}]-g(\mathbf{y})]|\mathbf{y}]\Big] \\
&= \mathbb{E}[\|\mathbf{x}-\mathbb{E}[\mathbf{x}|\mathbf{y}]\|_Q^2] + \mathbb{E}[\|\mathbb{E}[\mathbf{x}|\mathbf{y}]-g(\mathbf{y})\|_Q^2]
\end{aligned}
$$

where the last equality holds by taking the mean and since the second double product term is zero conditioning on $\mathbf{y}$. And since the two terms are non-negative and the first does not depend on g, we can minimize the second by setting $g(\mathbf{y}) = \mathbb{E}[\mathbf{x}|\mathbf{y}]$.

What if, instead of minimizing risk, one wants to maximize gain?

Bayesian inferece: MAP estimate

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Expected gain

$$G(z, y) := \mathbb{E}[(z - \mathbf{x})|\mathbf{y} = y] = \int_{\mathbb{R}^n} \gamma(z - x) f(x|y) dx$$

being γ a concave function with its max in the origin.

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By choosing γ to arbitrarily approximate Dirac's impulse $\delta \longrightarrow G(z, y) \approx f(z|y)$

The corresponding Max Expected Gain (or *Maximum a Posterior* - M.A.P.) estimator is given by

$$\hat{x}_{MAP}(y) := \text{Arg max}_z f(z|y)$$

which, for general $f(x|y)$ is the **conditional mode** ($\equiv \mathbb{E}[\mathbf{x}|\mathbf{y}]$ for unimodal symmetric densities).

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Unfortunately analytical computations of mean (or mode) are usually not possible.

Bayesian inference: the Gaussian case

The Gaussian case

If $\mathbf{x}$ and $\mathbf{y}$ are jointly Gaussian described by a density function with mean and variance

$$\mu_{\mathbf{z}} = \begin{bmatrix} \mu_{\mathbf{x}} \\ \mu_{\mathbf{y}} \end{bmatrix}; \quad \Sigma_{\mathbf{z}} = \begin{bmatrix} \Sigma_{\mathbf{x}} & \Sigma_{\mathbf{xy}} \\ \Sigma_{\mathbf{yx}} & \Sigma_{\mathbf{y}} \end{bmatrix}$$

the conditional density is still Gaussian with mean and variance (if $\Sigma_{\mathbf{y}} > 0^{a}$)

$$\mathbb{E}[\mathbf{x}|\mathbf{y}] = \mu_{\mathbf{x}} + \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}(\mathbf{y} - \mu_{\mathbf{y}})$$
$$\mathrm{Var}[\mathbf{x}|\mathbf{y}] = \Sigma_{\mathbf{x}} - \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}}$$

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- $\mathrm{Var}[\mathbf{x}|\mathbf{y}]$ (cond. var. of $\mathbf{x}|\mathbf{y}$) can be understood as $\Sigma_{\tilde{\mathbf{x}}}$ (uncond. var. of $\tilde{\mathbf{x}}$), does not depend on the data, and is given by the difference between $\Sigma_{\mathbf{x}}$ and $\mathrm{Var}\mathbb{E}[\mathbf{x}|\mathbf{y}] = \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}}$ (uncond. vars. of $\mathbf{x}$, $\mathbb{E}[\mathbf{x}|\mathbf{y}]$, resp.)

The Gaussian case: proof

Consider $\bar{\mathbf{z}} := \mathbf{z} - \mu_{\mathbf{z}} \sim \mathcal{N}(0, \Sigma_{\mathbf{z}})$. Consider a linear transformation of $\tilde{\mathbf{x}} = \bar{\mathbf{x}} + A\bar{\mathbf{y}}$, $\tilde{\mathbf{y}} = \bar{\mathbf{y}}$ such that $\tilde{\mathbf{x}} \perp \tilde{\mathbf{y}}$. That is

$$\mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{y}}^T] = 0 = \Sigma_{\mathbf{xy}} + A\Sigma_{\mathbf{y}} \implies A = -\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}$$

Clearly, since linear transforms maintain Gaussianity,

$$\tilde{\mathbf{z}} \sim \mathcal{N}(0, \Sigma_{\tilde{\mathbf{z}}}) = \begin{bmatrix} I & A \\ & I \end{bmatrix} \Sigma_{\mathbf{z}} \begin{bmatrix} I & \\ A^T & I \end{bmatrix} = \begin{bmatrix} \Sigma_{\tilde{x}} & \\ & \Sigma_{\mathbf{y}} \end{bmatrix}, \quad \Sigma_{\tilde{x}} = \Sigma_{\mathbf{x}} - \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}}$$

and being Gaussian and uncorrelated, $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ are also independent

$$\begin{aligned} \implies \bar{\mathbf{x}} = \tilde{\mathbf{x}} - A\bar{\mathbf{y}} \implies & \mathbb{E}[\bar{\mathbf{x}}|\bar{\mathbf{y}}] = -A\bar{\mathbf{y}} = \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\bar{\mathbf{y}} \\ & \mathrm{Var}[\bar{\mathbf{x}}|\bar{\mathbf{y}}] = \Sigma_{\tilde{\mathbf{x}}} = \Sigma_{\mathbf{x}} - \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}\Sigma_{\mathbf{yx}} \end{aligned}$$

Introducing the means you get the final result.

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- for $Q = I$, $\mathbb{E}\|\mathbf{x} - g(\mathbf{y})\|_Q^2$ is the scalar variance of the estimation error defined as $\tilde{\mathbf{x}} := \mathbf{x} - \mathbb{E}[\mathbf{x}|\mathbf{y}] \implies \mathbb{E}[\mathbf{x}|\mathbf{y}]$ is the *minimum variance estimator* of $\mathbf{x}$;

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What if the functional family of $g(\cdot)$ is a-priori restricted? E.g. to be linear

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Restricting $g(\cdot)$ to the family of linear unbiased functions of $\mathbf{y}$, we have

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Linear MMSE Estimator: the Problem (vector form)

The Linear MMSE Estimator is the linear function of the data

$$g(\mathbf{y}) = A\mathbf{y} + b, \quad A \in \mathbb{R}^{n \times m},\; b \in \mathbb{R}^n$$

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Before jumping to the solution...

Minimum introduction to Hilbert spaces

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$\Longrightarrow$ resorting to Hilbert spaces and, in particular, $\mathbf{H}(\mathbf{y})$ allows to interpret the problem geometrically!

Geometric formulation

Theorem: Orthogonal Projections (scalar form)

Let $\mathbf{x} \in \mathbf{H}$ be a (scalar) random variable. Let $\mathbf{H}(\mathbf{y}) \subset \mathbf{H}$. The variable $\mathbf{z}^* \in \mathbf{H}(\mathbf{y})$ minimizing $\|\mathbf{x} - \mathbf{z}\|^2_{\mathbf{H}} = \mathbb{E}[|\mathbf{x} - \mathbf{z}|^2]$ is unique and is the orthogonal projection of $\mathbf{x}$ onto $\mathbf{H}(\mathbf{y})$. Necessary and sufficient condition for $\mathbf{z}^*$ to be the orthogonal projection is that

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Thanks to this fundamental theorem, the properties inherited from Hilbert spaces and the correspondence between the notion of orthogonality ($\perp$) and uncorrelation (through the inner product/norm) a solution to the *Linear MMSE Estimator* can be drawn (directly in vector form from a element-wise application of the projection theorem).

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$$g(\mathbf{y}) = A\mathbf{y} + b, \quad A \in \mathbb{R}^{n \times m},\ b \in \mathbb{R}^n$$

that minimizes the mean of the squared estimation error $\mathbb{E}\|\mathbf{x} - g(\mathbf{y})\|^2$, that is

$$\min_{A,b} \mathbb{E}\|A\mathbf{y} + b - \mathbf{x}\|^2.$$

Linear MMSE Estimator: the Solution (vector form)

It can be shown that the solution $\hat{\mathbf{x}}(\mathbf{y}) = g^*(\mathbf{y})$ (also denoted $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$) is given by

$$\hat{\mathbf{x}}(\mathbf{y}) = \mu_{\mathbf{x}} + \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}(\mathbf{y} - \mu_{\mathbf{y}})$$

that is $A^* = \Sigma_{\mathbf{xy}}\Sigma_{\mathbf{y}}^{-1}$, $b^* = \mu_{\mathbf{x}} - A^*\mu_{\mathbf{y}}$. Equivalently to the Gaussian case, it depends only of the first and second moments of the joint density.

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- $\mathbb{E}[\mathbf{x}|\mathbf{y}]$ minimum variance estimator = conditional mean
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ linear minimum variance estimator $\neq$ conditional mean

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- $\mathbb{E}[\mathbf{x}|\mathbf{y}]$ minimum variance estimator $=$ conditional mean
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ linear minimum variance estimator $\neq$ conditional mean
- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ in general depends only on up to second order statistics

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- $\hat{\mathbb{E}}[\mathbf{x}|\mathbf{y}]$ very "far" from conditional mean for general densities

2. Linear Theory: Minimum Variance Linear Estimator for Stochastic Processes

Considerations

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1. Linear Estimators $\implies$ depending on first and second statistics, sufficient to find estimators minimizing squared measures of the estimation errors, e.g., variance.

2. Processes $\implies$ so far we considered $\mathbf{x}$ and $\mathbf{y}$ of fixed known dimensions. We now turn to the more interesting (and more useful) setting of "continuous" flow of new measurements $\{\mathbf{y}(t)\}$ of a (most often) time-varying signal $\{\mathbf{x}(t)\}$. In this scenario, the time interval I, during which data/measurements $\{\mathbf{y}(t)\}_{t\in I}$ flow, increases. Hence, it appears clearly how a suitable algorithmic structure to process an increasing amount of data becomes indeed fundamentally important.

Some more notation related to Hilber spaces...

- When dealing with (random) processes $\mathbf{y}(t)$ (collection of variables) over time, i.e., $\{\mathbf{y}_k(t), t \in \mathbb{Z}, k = \{1, \ldots, m\}\}$, the Hilbert space $\mathbf{H}(\mathbf{y}) \subset \mathbf{H}$ generated by the (second order) process can be defined as

$$\tilde{\mathbf{H}}(\mathbf{y}) = \left\{ \sum a_{ki} \mathbf{y}_k(t_i); a_{ki} \in \mathbb{R}, t_i \in \mathbb{Z}, k = \{1, \ldots, m\} \right\}$$

$$\mathbf{H}(\mathbf{y}) = \overline{\text{span}} \{\mathbf{y}(t), t \in \mathbb{Z}\} = \text{cls} \left\{ \tilde{\mathbf{H}}(\mathbf{y}) \right\}$$

with $\text{cls}\{\cdot\}$ denoting the formal operation of closure that is completing the corresponding space with the limits of all its Cauchy sequences which, in the case of infinite dimensional space, could not own to the original space.[4]

[4]We will take these technical aspects for granted.

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- Considering past and present time $s \in (-\infty, t]$, we denote

$$\mathbf{H}_t^-(\mathbf{y}) := \overline{\text{span}} \{\mathbf{y}(s); \ s \leq t\}; \quad \mathbf{H}_s^-(\mathbf{y}) \subseteq \mathbf{H}_t^-(\mathbf{y}), s \leq t; \quad \lim_{t \to \infty} \mathbf{H}_t^-(\mathbf{y}) = \mathbf{H}(\mathbf{y})$$

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- Considering present and future time $s \in [t, +\infty)$, we denote

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In particular if $s < t$

$$\hat{\mathbb{E}}\left[\hat{\mathbb{E}}[\mathbf{x}|\mathbf{H}_t^-(\mathbf{y})]|\mathbf{H}_s^-(\mathbf{y})\right] = \hat{\mathbb{E}}[\mathbf{x}|\mathbf{H}_s^-(\mathbf{y})]$$

State Space models

Let $\{\mathbf{y}(t)\}$ an m-dimensional stochastic process defined on $[t_0, \infty)$.

[5] To avoid notation clutter, time dependence of the model matrices is omitted.

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Let $\{\mathbf{y}(t)\}$ an m-dimensional stochastic process defined on $[t_0, \infty)$. We say it is a *finite dimensional process* if we can express it as

$$\begin{aligned} \mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{n}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{n}(t), \quad t \geq t_0, \ \mathbf{x}(t_0) = \mathbf{x}_0 \end{aligned}$$

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- $\{\mathbf{x}(t)\}$ is a n-dimensional process, called *state process*, with initial value $\mathbf{x}_0$ which is uncorrelated w.r.t. all present and future values of $\mathbf{n}$, i.e., $\mathbb{E}[x_0\mathbf{n}(t)^T] = 0$, $\forall t \geq t_0$ and $\mathbb{E}[\mathbf{x}_0] = \mu_0,\ Var[\mathbf{x}_0] = \Sigma_0$

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Essentially a finite dimensional process can be described as the output of a (time-varying) linear dynamical system driven by white noise.

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State Space models - the state process

Markov property

$\{\mathbf{x}(t)\}$ is a weak (sometimes also referred to as second-oder) Markov process, i.e.,

$$\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s(\mathbf{x})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)] \quad \forall t \geq s$$

being $\mathbf{H}_s(\mathbf{x}) = \mathbf{H}_s^-(\mathbf{x})$ the Hilbert space generated by $\{\mathbf{x}(t_0), \ldots, \mathbf{x}(s)\}$. If $\{\mathbf{n}(t)\}$ and $\mathbf{x}_0$ are jointly Gaussian, $\{\mathbf{x}(t)\}$ is strongly Markov (i.e., replace $\hat{\mathbb{E}}$ by $\mathbb{E}$, equivalently the entire distribution).

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Sketch of proof:

$$\mathbf{x}(t) = A^{t-t_0}\mathbf{x}_0 + \sum_{k=t_0}^{t-1} A^{t-1-k}B\mathbf{n}(k) = A^{t-s}\mathbf{x}(s) + \sum_{k=s}^{t-1} A^{t-1-k}B\mathbf{n}(k), \quad t \geq s$$

$$\Rightarrow \{\mathbf{x}(t)\}\text{linear in } \mathbf{x}_0,\ \{\mathbf{n}(t_0), \ldots, \mathbf{n}(t-1)\} \Rightarrow \mathbf{x}(t) \perp \mathbf{H}_t^+(\mathbf{n}) \Rightarrow \mathbf{H}_s^-(\mathbf{x}) \perp \mathbf{H}_s^+(\mathbf{n})$$

$$\Rightarrow \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s^-(\mathbf{x})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)] = A^{t-s}\mathbf{x}(s)$$

For the Gaussian case, $\hat{\mathbb{E}} \equiv \mathbb{E}$. Also the entire conditional probability depends on the past only through the conditional mean hence the property holds "globally".

State Space models - the state process

The same "memory" property holds wrt the output process $\{\mathbf{y}(t)\}$

$$\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_s^-(\mathbf{x},\mathbf{y})] = \hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{x}(s)]$$
$$\hat{\mathbb{E}}[\mathbf{y}(t)|\mathbf{H}_s^-(\mathbf{x},\mathbf{y})] = \hat{\mathbb{E}}[\mathbf{y}(t)|\mathbf{x}(s)], \quad t \geq s$$

being $\mathbf{H}_s^-(\mathbf{x},\mathbf{y}) = \mathbf{H}_s^-(\mathbf{x}) \vee \mathbf{H}_{s-1}^-(\mathbf{y})$ vector sum of $\mathbf{H}_s^-(\mathbf{x})$ (past and present wrt to s) and $\mathbf{H}_{s-1}^-(\mathbf{y})$ (strict past wrt to s).

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Conditional Incorrelation (Orthogonality)

Define $\mathbf{z}(t) := [\mathbf{x}(t+1)^T, \mathbf{y}(t)^T]^T$,

$$\mathbf{H}_t^-(\mathbf{z}) := \mathbf{H}_{t+1}^-(\mathbf{x}) \vee \mathbf{H}_t^-(\mathbf{y}), \quad \mathbf{H}_t^+(\mathbf{z}) := \mathbf{H}_{t+1}^+(\mathbf{x}) \vee \mathbf{H}_t^+(\mathbf{y})$$
$$\mathbf{X}_t := \text{span}\{\mathbf{x}_1(t), \ldots, \mathbf{x}_n(t)\} \quad \text{state space of } \mathbf{x} \text{ at t}$$

then

$$\mathbf{H}_t^-(\mathbf{z}) \perp \mathbf{H}_t^+(\mathbf{z}) \mid \mathbf{X}_t \implies \hat{\mathbf{z}}(t) = \hat{\mathbb{E}}[\mathbf{z}(t)|\mathbf{x}(t)] = \begin{bmatrix} A \\ C \end{bmatrix} \mathbf{x}(t)$$

State Space models - moments

From $\mathbb{E}[\mathbf{n}] = 0$ and previous (conditional) incorrelation properties, the evolution of mean and covariance of a linear finite dimensional system is described by

$$\begin{aligned}
\mu_{\mathbf{x}}(t+1) &= A\mu_{\mathbf{x}}(t), \quad \mu_{\mathbf{x}}(t_0) = \mu_0 \\
\mu_{\mathbf{y}}(t) &= C\mu_{\mathbf{x}}(t) \\
\Sigma_{\mathbf{x}}(t,s) &= \begin{cases} A^{t-s}\Sigma(s), & t \geq s \\ \Sigma(t)(A^T)^{s-t}, & t \leq s \end{cases} \\
\Sigma_{\mathbf{y}}(t,s) &= \begin{cases} CA^{t-s-1}G(s), & t > s \\ C\Sigma(t)C^T + DD^T, & t = s \\ G(t)^T(A^T)^{s-t-1}C^T, & t < s \end{cases} \\
G(s) &= A\Sigma(s)C^T + BD^T
\end{aligned}$$

and $\Sigma(t) = \mathrm{Var}[\mathbf{x}(t)]$ is the solution of the discrete Lyapunov equation

$$\Sigma(t+1) = A\Sigma(t)A^T + BB^T, \quad \Sigma(t_0) = \Sigma_0$$

State Space models - stability and (asymptotic) stationarity

Stability and (asymptotic) stationarity[6] of the underlying processes play an important role in the relation between the Wiener-Kolmogorov and Kalman theories.

[6] Evolution of covariance depends on $\tau = t - s$, $\Sigma(s, t) = \Sigma(\tau)$, for $t, s \geq t_0$ and $t - t_0 \to \infty$.

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- If $\mathbf{x}_0 \sim (0, \Sigma_\infty)$, then $\{\mathbf{x}(t)\}, \{\mathbf{y}(t)\}$ are weakly stationary ($\forall t \geq t_0$) with

$$\Sigma_{\mathbf{y}} = C\Sigma_\infty C^T + DD^T.$$

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- $\Sigma(t+1) = A\Sigma(t)A^T + BB^T \to \Sigma_\infty = \sum_{k=0}^{\infty} A^k BB^T (A^T)^k$ (unique and $\succ 0$)
- If $\mathbf{x}_0 \sim (0, \Sigma_\infty)$, then $\{\mathbf{x}(t)\}, \{\mathbf{y}(t)\}$ are weakly stationary ($\forall t \geq t_0$) with

$$\Sigma_{\mathbf{y}} = C\Sigma_\infty C^T + DD^T.$$

Under the above conditions[a] the two approaches (Wiener-Kolmogorov and Kalman) are essentially equivalent.

[a] For finite LTI systems, stationarity translates into rationality of the transfer function and, in turn, ARMA/ARMAX representation. Hence the possibility to practically compute the filter transfer function.

[6] Evolution of covariance depends on $\tau = t - s$, $\Sigma(s,t) = \Sigma(\tau)$, for $t, s \geq t_0$ and $t - t_0 \to \infty$.

State Space models - a common alternative notation

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\begin{aligned}
\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{n}(t), \quad \mathbb{E}[\mathbf{n}(t)] = 0,\ \mathbb{E}[\mathbf{n}(t)\mathbf{n}(s)^T] = I\delta_{ts} \\
\mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{n}(t), \quad t \geq t_0,\ \mathbf{x}(t_0) = \mathbf{x}_0
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By defining $\mathbf{v}(t) := B\mathbf{n}(t)$ (state/process noise) and $\mathbf{w}(t) := D\mathbf{n}(t)$ (measurement/observation noise) and $\mathbf{z}(t) = \begin{bmatrix} \mathbf{x}(t)^T & \mathbf{y}(t-1)^T \end{bmatrix}^T$, we have

$$\mathbf{z}(t+1) = \begin{bmatrix} A \\ C \end{bmatrix} \mathbf{z}(t) + \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix}, \quad \mathbb{E}\begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = 0,\ \mathrm{Var}\begin{bmatrix} \mathbf{v}(t) \\ \mathbf{w}(t) \end{bmatrix} = \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix}$$

Often, in practical scenarios, $S = 0$ i.e. uncorrelated process and measurement noise.

(Some) Realization techniques: stationary processes

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Assumptions (for simplicity):

- Scalar (input and output, SISO) processes. For MIMO processes, generalization using matrix polynomials
- $A(z^{-1})$ and $C(z^{-1})$ are already monic (highest order coeff. $=1$)

Realization: Controllable Canonical Form

The ARMAX description in transfer functions terms can be written:

$$y_k = \frac{B(z^{-1})}{A(z^{-1})} u_k + \frac{C(z^{-1})}{A(z^{-1})} e_k$$

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Controllable form: controllability matrix from (A_c, B_c) has full rank, i.e., the system is controllable (input drives the state everywhere in finite time).

Realization: Controllable Form - Example (part 1)

Consider the ARMAX model

$$y_k = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} u_k + \frac{1 + c_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} e_k$$

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$$A_c = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix}, \; B_c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; D_u = b_0, \; D_e = 1,$$
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corresponding to the transfer function

$$y_k = (C_u (zI - A_c)^{-1} B_c + D_u) u_k + (C_e (zI - A_c)^{-1} B_c + D_e) e_k$$

Realization: Controllable Form - Example (part 2)

Explicit computation lead (for the first term, the second can be similarly checked)

$$
\begin{aligned}
C_u(zI - A_c)^{-1}B_c + D_u &= \begin{bmatrix} b_1 - a_1 b_0 & -a_2 b_0 \end{bmatrix} \begin{bmatrix} z + a_1 & a_2 \\ -1 & z \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b_0 \\
&= \begin{bmatrix} b_1 - a_1 b_0 & -a_2 b_0 \end{bmatrix} \frac{1}{z^2 + a_1 z + a_2} \begin{bmatrix} z & -a_2 \\ 1 & z + a_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b_0 \\
&= \frac{(b_1 - a_1 b_0)z - a_2 b_0 + b_0(z^2 + a_1 z + a_2)}{z^2 + a_1 z + a_2} \\
&= \frac{b_1 z + b_0 z^2}{z^2 + a_1 z + a_2} \\
&= \frac{z}{z^2} \frac{b_1 + b_0 z}{1 + a_1 z^{-1} + a_2 z^{-2}} \\
&= z^{-1} \frac{b_1 + b_0 z}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}}
\end{aligned}
$$

Realization: Observable Canonical Form (Dual)

A dual (observable) state space form with state $x_k \in \mathbb{R}^{n_a}$ is

$$x_{k+1} = A_o\, x_k + B_u\, u_k + B_e\, e_k, \qquad y_k = C_o\, x_k + D_u\, u_k + D_e\, e_k.$$

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$$A_o = A_c^\top = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad C_o = B_c^\top = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}$$

$$B_u := \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix}, \quad B_e := \begin{bmatrix} c_1 - a_1 \\ c_2 - a_2 \\ \vdots \\ c_n - a_n \end{bmatrix} \quad D_u = b_0, \quad D_e = 1$$

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Observable form: observability matrix from (A_o, C_o) has full rank, i.e., the state can be reconstructed from a finite window of outputs.

Realization: Innovation (Kalman) Form

An **innovation-form** (a.k.a. **Kalman form**) state-space model is

$$x_{k+1} = A\,x_k + B\,u_k + K\,e_k, \qquad y_k = C\,x_k + D\,u_k + e_k.$$

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For monic ARMAX, a convenient choice, consistent with the controllable realization, is

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Note that in the above SS model there is some abuse of notation since to interpret e_k as the innovation the model does not define the evolution of the "true" state x_k but rather of the prediction $\hat{x}_k$.

Realization: Observer (Predictor) Form

Assuming an innovation-form model

$$x_{k+1} = A\,x_k + B\,u_k + K\,e_k, \qquad y_k = C\,x_k + D\,u_k + e_k.$$

Eliminate the unmeasured innovation e_k using $e_k = y_k - (C\,x_k + D\,u_k)$.

Substitute into the state update to get the **observer / predictor form**:

$$x_{k+1} = (A - KC)\,x_k + (B - KD)\,u_k + K\,y_k, \qquad \hat{y}_{k|k-1} = C\,x_k + D\,u_k.$$

For consistency with the controllable ARMAX realization:

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Observer (predictor) form: state propagated using the measured output y_k.

Note: this form is not really generative. Rather y_k are assumed to be available.

Discretization: CT to DT (Exact) (part 1)

In some case, the SS model is already available but it is in CT. Hence one might want to discretize it.

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Consider a general CT model

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Assume a sampling period $T > 0$ and **zero-order hold (ZOH)** input, the equations for $t \in [kT, (k+1)T]$ leads to

$$x(t) = e^{A(t-kT)}x_k + \int_{kT}^{t} e^{A(t-\tau)}B\,u(\tau)\,d\tau + \int_{kT}^{t} e^{A(t-\tau)}G\,w(\tau)\,d\tau.$$

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and evaluate at $t = (k+1)T$ to

$$x_{k+1} = e^{AT}x_k + \left(\int_{0}^{T} e^{A(T-\tau)}B\,d\tau\right)u_k + \int_{0}^{T} e^{A(T-\tau)}G\,w(kT+\tau)\,d\tau.$$

Discretization: CT to DT (Exact) (part 2)

With the change of variable $\sigma = T - \tau$ and output sampling, we get:

$$x_{k+1} = A_d x_k + B_d u_k + w_k$$
$$y_k = C x_k + D u_k + v_k.$$

$$A_d := e^{AT}, \quad B_d := \int_0^T e^{A\sigma} B \, d\sigma, \quad w_k := \int_0^T e^{A\sigma} G \, w(kT + T - \sigma) \, d\sigma.$$

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If $w(t)$ is CT white noise with spectral density Q_c, $E[w(t)w(\tau)^\top] = Q_c \, \delta(t - \tau)$, then w_k is discrete-time white with

$$Q_d := E[w_k w_k^\top] = \int_0^T e^{A\sigma} \, G \, Q_c \, G^\top \, e^{A^\top \sigma} \, d\sigma.$$

Note: there are also other (simpler but inexact) discretization techniques, e.g. Forward/Backward Euler etc.

Example of non-stationary process

A process is **stationary** when its statistical properties (mean, variance, autocorrelation) do not change over time. Typically arise from **stable systems**. For ARMAX representations it essentially means that the poles of the transfer function (i.e. the root of $A(z^{-1})$) being inside the unit circle. Examples are:

- thermal noise in an electrical resistor
- vibrations in mechanical systems
- turbulent flow velocity fluctuations
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A process is **non-stationary** when its statistical properties change over time. Often caused by: integration, accumulation, system drift.

- particle under Brownian motion (diffusion process)
- position of a vehicle (tracking navigation)
- financial asset prices

Example of non-stationary process

A process is **stationary** when its statistical properties (mean, variance, autocorrelation) do not change over time. Typically arise from **stable systems**. For ARMAX representations it essentially means that the poles of the transfer function (i.e. the root of $A(z^{-1})$) being inside the unit circle. Examples are:

- thermal noise in an electrical resistor
- vibrations in mechanical systems
- turbulent flow velocity fluctuations
- noise in sensors

A process is **non-stationary** when its statistical properties change over time. Often caused by: integration, accumulation, system drift.

- particle under Brownian motion (diffusion process)
- position of a vehicle (tracking navigation)
- financial asset prices

$\Longrightarrow$ Ofter captured/modeles using ARIMA models.

ARIMA(p, d, q)

Let $\Delta := 1 - z^{-1}$ (difference operator). An ARIMA(p, d, q) process satisfies

$$A(z^{-1})\,\Delta^d y_k = C(z^{-1})\,e_k, \qquad e_k \sim WN(0, \sigma^2),$$

with monic polynomials

$$A(z^{-1}) = 1 + a_1 z^{-1} + \cdots + a_p z^{-p}, \qquad C(z^{-1}) = 1 + c_1 z^{-1} + \cdots + c_q z^{-q}.$$

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For an ARIMA model, define the stationary differenced series

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Note that algebraically ARIMA(p,d,q)=ARMA(p+d,q). However, the d roots are now on the unit circle due to the integrators which would mean "non-stationary" ARMA which in turn does not make to much sense in standard time-series terminology.

ARIMA(p, d, q): State-Space Derivation

Find a n-dim realization ($n = \max\{p, q+1\}$) for the ARMA for z_k (e.g. in innovation form)

$$x_{k+1}^z = A_z\, x_k^z + K_z\, e_k, \qquad z_k = C_z\, x_k^z + e_k,$$

where (A_z, C_z) are chosen so that the transfer $e \mapsto z$ equals $\frac{C(q^{-1})}{A(q^{-1})}$.

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Add d discrete integrators $s^{(1)}_k, \ldots, s^{(d)}_k$ to recover $y_k = s^{(d)}_k$ from $z_k = \Delta^d y_k$

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$$s_{k+1} = T_d\, s_k + b_d\, z_{k+1}, \quad y_k = e_d^\top s_k, \quad b_d^\top = \begin{bmatrix}1, 1, \dots, 1\end{bmatrix}, \; e_d^\top = \begin{bmatrix}0, \dots, 0, 1\end{bmatrix}$$

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Let be $x_k := \begin{bmatrix} (x^z_k)^\top & s_k^\top \end{bmatrix}^\top$, using the ARMA equations

$$x_{k+1} = \begin{bmatrix} A_z & 0 \\ b_d\, C_z A_z & T_d \end{bmatrix} x_k + \begin{bmatrix} K_z \\ b_d\, C_z K_z \end{bmatrix} e_k + \begin{bmatrix} 0 \\ b_d \end{bmatrix} e_{k+1},$$

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ARIMA(1,2,1) Example (part 1)

Consider the following ARIMA(1,2,1)

$$(1 + a_1 z^{-1})(1 - z^{-1})^2 y_k = (1 + c_1 z^{-1}) e_k$$

where $\Delta^2 = (1 - z^{-1})^2 = 1 - 2z^{-1} + z^{-2}$.

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We define $z_k = \Delta^2 y_k$ and derive a SS representation for its ARMA(1,1) model. E.g. controllable companion form that is

$$x^z_{k+1} = -a_1 x^z_k + e_k, \quad z_k = (c_1 - a_1) x^z_k + e_k$$

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$$s_{k+1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} s_k + \begin{bmatrix} 1 \\ 1 \end{bmatrix} z_{k+1}, \quad y_k = \begin{bmatrix} 0 & 1 \end{bmatrix} s_k$$

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Combining the models ($x_k = [x^z_k, s_k^{(1)}, s_k^{(2)}]^T$) we finally get

$$x_{k+1} = \begin{bmatrix} -a_1 & 0 \\ -\begin{bmatrix} 1 \\ 1 \end{bmatrix} (c_1 - a_1) a_1 & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{bmatrix} x_k + \begin{bmatrix} 1 \\ \begin{bmatrix} 1 \\ 1 \end{bmatrix} (c_1 - a_1) \end{bmatrix} e_k + \begin{bmatrix} 0 \\ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{bmatrix} e_{k+1},$$

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One could investigate if the SS model does indeed correspond to the original ARIMA(1,2,1) model, e.g., checking the transfer function $G(z)$ for I/O behavior.

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$$G(z) = \begin{bmatrix} 0 & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{bmatrix} \underbrace{\left(zI - \begin{bmatrix} -a_1 & 0 \\ -\begin{bmatrix} 1 \\ 1 \end{bmatrix} (c_1 - a_1) a_1 & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{bmatrix} \right)^{-1}}_{\begin{bmatrix} * & * & * \\ * & * & * \\ -\frac{a_1 z (c_1 - a_1)}{(z+a_1)(z-1)^2} & \frac{1}{(z-1)^2} & \frac{1}{z-1} \end{bmatrix}} \left(z \begin{bmatrix} 0 \\ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{bmatrix} + \begin{bmatrix} 1 \\ \begin{bmatrix} 1 \\ 1 \end{bmatrix} (c_1 - a_1) \end{bmatrix} \right)$$

$$= -\frac{a_1 z (c_1 - a_1)}{(z + a_1)(z-1)^2} + \frac{z + c_1 - a_1}{(z-1)^2} + \frac{z + c_1 - a_1}{z - 1}$$

$$= \frac{z^2 (z + c_1)}{(z + a_1)(z - 1)^2}$$

$$= \frac{1 + c_1 z^{-1}}{(1 + a_1 z^{-1})(1 - z^{-1})^2} \quad \Longrightarrow \text{ The original ARIMA}$$

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Consider for instance a very simple ARIMA$(0, 2, 0)$ model such as

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where $e_k \sim WN(0, Q)$, $Q = T^2 \sigma_a^2 I_2$ and measurements

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That is, the above model is equivalent to

$$\begin{aligned} p_{k+1} &= p_k + T v_k, \\ v_{k+1} &= v_k + T a_k, \qquad a_k \sim WN(0, \sigma_a^2 I_2) \end{aligned}$$

where now $p = x$, v and $a = e$ are position, velocity and acceleration respectively.

The Kalman Filter: reference state space model

In the following we consider the general state space model of the process $\{\mathbf{y}(t)\}$

$$
\begin{aligned}
\mathbf{x}(t+1) &= A\mathbf{x}(t) + \mathbf{v}(t) \\
\mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{w}(t) \quad \mathbb{E}\begin{bmatrix}\mathbf{v}(t)\\ \mathbf{w}(t)\end{bmatrix} = 0,\ \operatorname{Var}\begin{bmatrix}\mathbf{v}(t)\\ \mathbf{w}(t)\end{bmatrix} = \begin{bmatrix} Q & S \\ S^T & R\end{bmatrix}, R \succ 0 \\
\mathbf{x}(t_0) &= \mathbf{x}_0, \quad \mathbb{E}[\mathbf{x}_0] = \mu_0, \quad \operatorname{Var}[\mathbf{x}_0] = P_0 \succcurlyeq 0, \quad \mathbf{x}_0 \perp \mathbf{v}(t), \mathbf{w}(t)\ \forall t \geq t_0
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Note: A, C, Q, S, R could be time varying. $R \succ 0$ for simplicity for now because it helps guaranteeing invertibility of the variance of innovation process $\mathbf{e}(t) = \mathbf{y}(t) - \hat{\mathbf{y}}(t|t-1)$.

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$$\mathbf{x}(t+1) = \begin{bmatrix} \mathbf{x}_1(t+1) \\ \mathbf{x}_2(t+1) \end{bmatrix} = \begin{bmatrix} A_1 & \\ & A_2 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} = A\mathbf{x}(t) + \mathbf{v}(t)$$

$$\mathbf{y}(t) = \mathbf{z}(t) + \xi(t) = \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \begin{bmatrix} I & I \end{bmatrix} \begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix} = C\mathbf{x}(t) + \mathbf{w}(t)$$

$$\mathbf{z}(t) = \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \end{bmatrix} + \mathbf{w}_1(t) = \overline{C}_1\mathbf{x}(t) + \mathbf{w}_1(t)$$

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But what about $\hat{\mathbf{w}}_1(t|t)$?

$$\mathbf{w}(t) \not\perp \mathbf{w}_1(t) \text{ with } \mathbb{E}[\mathbf{w}_1(t)\mathbf{w}(t)^\top] = R_1 \Longrightarrow \mathbf{y}(t) \not\perp \mathbf{w}_1(t)$$

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Again, the filtering solution can be directly built for the filtered state!

The Kalman Filter: reference state space model (modified)

$$\mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{v}(t)$$

$$\mathbf{y}(t) = C\mathbf{x}(t) + \mathbf{w}(t) \quad \mathbb{E}\begin{bmatrix}\mathbf{v}(t)\\ \mathbf{w}(t)\end{bmatrix} = 0,\ \mathrm{Var}\begin{bmatrix}\mathbf{v}(t)\\ \mathbf{w}(t)\end{bmatrix} = \begin{bmatrix}Q & S\\ S^T & R\end{bmatrix}, R \succ 0$$

$$\mathbf{x}(t_0) = \mathbf{x}_0, \quad \mathbb{E}[\mathbf{x}_0] = \mu_0, \quad \mathrm{Var}[\mathbf{x}_0] = P_0 \succcurlyeq 0, \quad \mathbf{x}_0 \perp \mathbf{v}(t), \mathbf{w}(t)\ \forall t \geq t_0$$

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$$\implies \mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{v}(t) = F\mathbf{x}(t) + SR^{-1}\mathbf{y}(t) + \tilde{\mathbf{v}}(t), \quad F = A - SR^{-1}C$$

with uncorrelated noise $\tilde{\mathbf{v}}(t) \perp \mathbf{w}(t)$.

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1. A-priori estimates/predictions (temporal update)
   - State: $\hat{\mathbf{x}}(t+1|t) = F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t)$
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where

- $P(t+1|t)$, $P(t|t)$: the prediction and filtering **errors** variances
- $\Lambda(t) = CP(t|t-1)C^T + R$: innovation ($\mathbf{e}(t) = \mathbf{y}(t) - C\hat{\mathbf{x}}(t|t-1)$) variance
- $L(t) = P(t|t-1)C^T\Lambda(t)^{-1}$: Kalman gain

The Kalman Filter: equations - proof (part 1)

1. **A-priori (prediction) update.** The one-step ahead estimate is obtained by taking the linear conditional expectation given past measurements $\mathbf{H}_t^-(\mathbf{y})$, recalling that $\tilde{\mathbf{v}}(t) \perp \mathbf{w}(t) \implies \tilde{\mathbf{v}}(t) \perp \mathbf{H}_t^-(\mathbf{y})$:

$$\hat{\mathbf{x}}(t+1|t) := \hat{\mathbb{E}}[\mathbf{x}(t+1)|\mathbf{H}_t^-(\mathbf{y})] = F\hat{\mathbb{E}}[\mathbf{x}(t)|\mathbf{H}_t^-(\mathbf{y})] + SR^{-1}\mathbf{y}(t) = F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t)$$

The corresponding prediction error $\tilde{\mathbf{x}}(t+1|t) = \mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t)$ satisfies

$$\tilde{\mathbf{x}}(t+1|t) = F\tilde{\mathbf{x}}(t|t) + \tilde{\mathbf{v}}(t)$$

and, since $\tilde{\mathbf{v}}(t) \perp \tilde{\mathbf{x}}(t|t)$ because

$$\left.\begin{array}{l} \mathbf{x}(t) \in \text{span}\,\{\mathbf{x}(t_0), \tilde{\mathbf{v}}(s), \mathbf{w}(s);\ \ s < t\} \\ \mathbf{H}_t^-(\mathbf{y}) \subset \text{span}\,\{\mathbf{x}(t_0), \tilde{\mathbf{v}}(s-1), \mathbf{w}(s); s \leq t\} \end{array}\right\} \perp \tilde{\mathbf{v}}(t)$$

its covariance evolves as

$$P(t+1|t) = \text{Var}[\tilde{\mathbf{x}}(t+1|t)] = FP(t|t)F^T + \tilde{Q}$$

2. **A-posteriori (measurement) update.** At time $t+1$ we get $\mathbf{y}(t+1) = C\mathbf{x}(t+1) + \mathbf{w}(t+1)$. Note that

$$\mathbf{H}_{t+1}^{-}(\mathbf{y}) = \mathbf{H}_{t}^{-}(\mathbf{y}) \oplus \mathbf{H}(\mathbf{e}(t+1)) \quad \text{(orthogonal sum)}$$

being

$$\mathbf{e}(t+1) = \mathbf{y}(t+1) - C\hat{\mathbf{x}}(t+1|t) = C\tilde{\mathbf{x}}(t+1|t) + \mathbf{w}(t+1)$$

$$\begin{aligned}\Longrightarrow \hat{\mathbf{x}}(t+1|t+1) = \hat{\mathbb{E}}[\mathbf{x}(t+1)|\mathbf{H}_{t+1}^{-}(\mathbf{y})] &= \hat{\mathbf{x}}(t+1|t) + L(t+1)\mathbf{e}(t+1) \\ &= \hat{\mathbf{x}}(t+1|t) + L(t+1)\big(\mathbf{y}(t+1) - C\hat{\mathbf{x}}(t+1|t)\big)\end{aligned}$$

where $L(t+1) = \mathrm{Cov}[\mathbf{x}(t+1), \mathbf{e}(t+1)]\mathrm{Var}[\mathbf{e}(t+1)]^{-1}$ with

$$\begin{aligned}\mathrm{Cov}[\mathbf{x}(t+1), \mathbf{e}(t+1)] &= \mathrm{Cov}[\mathbf{x}(t+1), \tilde{\mathbf{x}}(t+1|t)]C^T + \mathrm{Cov}[\mathbf{x}(t+1), \mathbf{w}(t+1)] \\ &= \mathrm{Cov}[\tilde{\mathbf{x}}(t+1|t), \tilde{\mathbf{x}}(t+1|t)]C^T = P(t+1|t)C^T \\ \mathrm{Var}[\mathbf{e}(t+1)] = \Lambda(t+1) &= CP(t+1|t)C^T + R, \quad \text{since } \tilde{\mathbf{x}}(t+1|t) \perp \mathbf{w}(t+1)\end{aligned}$$

The Kalman Filter: equations - proof (part 3)

The covariance update follows by noting that

$$
\begin{aligned}
&\mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t+1) = \mathbf{x}(t+1) - \hat{\mathbf{x}}(t+1|t) - L(t+1)\mathbf{e}(t+1) \\
&\Longrightarrow \tilde{\mathbf{x}}(t+1|t+1) = \tilde{\mathbf{x}}(t+1|t) - L(t+1)\mathbf{e}(t+1) \\
&\Longrightarrow \tilde{\mathbf{x}}(t+1|t+1) + L(t+1)\mathbf{e}(t+1) = \tilde{\mathbf{x}}(t+1|t)
\end{aligned}
$$

and being $\tilde{\mathbf{x}}(t+1|t+1) \perp \mathbf{e}(t+1)$

$$
P(t+1|t+1) + L(t+1)\Lambda(t+1)L(t+1)^T = P(t+1|t)
$$

$$
\begin{aligned}
\Longrightarrow P(t+1|t+1) &= P(t+1|t) - L(t+1)\Lambda(t+1)L(t+1)^T \\
&= P(t+1|t) - P(t+1|t)C^T\Lambda(t+1)^{-1}CP(t+1|t)
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Considerations

1. To avoid/reduce numerical ill-conditioning it is sometimes preferred to express the covariance update in different forms

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\begin{aligned}
P(t+1|t+1) &= P(t+1|t) - P(t+1|t)C^T\Lambda(t+1)^{-1}CP(t+1|t) \\
&= (I - L(t+1)C)\,P(t+1|t) \\
\text{symmetrized form} &= [I - L(t+1)C]\,P(t+1|t)\,[I - L(t+1)C]^T + \\
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4. So far, with no additional assumptions on the specific distributions of $\mathbf{w}, \mathbf{v}, \mathbf{x}_0$, KF is optimal among linear estimators $\hat{\mathbb{E}}[\cdot]$. Assuming joint Gaussianity, KF is the true MMSE/Bayesan estimator ($\hat{\mathbb{E}} \equiv \mathbb{E}$).

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It is often useful to express the dynamics of the predictor which allows to express the next state prediction as function of the previous without computing the filtered value. That is (after some manipulation)

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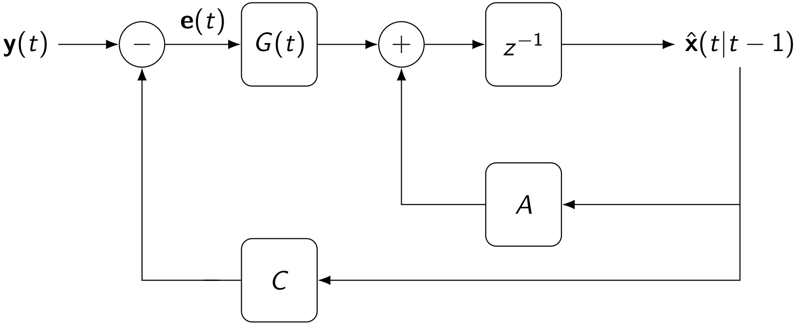
\begin{aligned}
\hat{\mathbf{x}}(t+1|t) &= F(I - L(t)C)\hat{\mathbf{x}}(t|t-1) + (FL(t) + SR^{-1})\mathbf{y}(t) \\
&= (F - K(t)C)\hat{\mathbf{x}}(t|t-1) + (K(t) + SR^{-1})\mathbf{y}(t) \\
&= \Gamma(t)\hat{\mathbf{x}}(t|t-1) + G(t)\mathbf{y}(t) \\
&= A\hat{\mathbf{x}}(t|t-1) + G(t)\left[\mathbf{y}(t) - C\hat{\mathbf{x}}(t|t-1)\right] \\
&= A\hat{\mathbf{x}}(t|t-1) + G(t)\mathbf{e}(t) \\
P(t+1|t) &= F\left[P(t|t-1) - P(t|t-1)C^T\Lambda(t)^{-1}CP(t|t-1)\right]F^T + \tilde{Q} \\
&= F\left[I - L(t)C\right]P(t|t-1)\left[I - L(t)C\right]^T F^T + FL(t)RL^T(t)F^T + \tilde{Q}
\end{aligned}
$$

where $K(t) := FL(t)$, $\Gamma(t) := F - K(t)C$, $G(t) := K(t) + SR^{-1}$.

Equivalent closed loop interpretation



where the closed-loop dynamics is described by

$$\Gamma(t) = A - G(t)C = F - K(t)C$$

Hence its properties characterize the filter/predictor dynamics.

Filtered Estimates

Similarly (for $S = 0$ for simplicity)

$$\begin{aligned}
\hat{\mathbf{x}}(t+1|t+1) &= A\hat{\mathbf{x}}(t|t) + L(t+1)\left[\mathbf{y}(t+1) - CA\hat{\mathbf{x}}(t|t)\right] \\
P(t+1|t+1) &= \left[I - L(t+1)C\right]P(t+1|t)
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In this case the closed-loop dynamical behavior is described by

$$\Gamma_{\text{filt}}(t) = \left[I - L(t+1)C\right]A$$

where before, with $S = 0$, was $\Gamma_{\text{pred}}(t) = A\left[I - L(t)C\right]$ which suggest an almost identical asymptotic dynamic behavior.

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Note that prediction and filtered estimates can directly be computed from the original equations. It is sometimes convenient to get the explicit expressions in order to better understand the role of the measurements in the update.

Kalman Filter with Input (part 1)

Reference model (with usual assumptions on noise and initial state):

$$\begin{aligned}\mathbf{x}(t+1) &= A\mathbf{x}(t) + B\mathbf{u}(t) + \mathbf{v}(t) = F\mathbf{x}(t) + SR^{-1}\mathbf{y}(t) + B\mathbf{u}(t) + \tilde{\mathbf{v}}(t)\\ \mathbf{y}(t) &= C\mathbf{x}(t) + \mathbf{w}(t) + D\mathbf{u}(t)\end{aligned}$$

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Easy: if $\mathbf{u}(t)$ **known and deterministic**

$$\begin{aligned}\hat{\mathbf{x}}(t+1|t) &= F\hat{\mathbf{x}}(t|t) + SR^{-1}\mathbf{y}(t) + B\mathbf{u}(t)\\ P(t+1|t) &= \dots\ \textit{unchanged}\\ \mathbf{e}(t) &= y(t) - C\hat{\mathbf{x}}(t|t-1) - D\mathbf{u}(t)\end{aligned}$$

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Harder: *but what if* $\mathbf{u}(t)$ *is a process?* E.g. a (linear) causal function (H) of $\mathbf{H}_t^-(\mathbf{y})$ plus a process uncorrelated from the entire noise $(\mathbf{v}, \mathbf{w})$ history, say

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In this case, even if $H \equiv 0$, it is in general not possible to assume $\mathbf{u}(t) \perp \mathbf{x}_0$. Even if $\mathbf{u}(t)$ is observable, $\hat{\mathbf{x}}$ based on joint observations $(\mathbf{y}, \mathbf{u})$ must be obtained by considering the joint state model thus deriving augmented KF equations.

Kalman Filter with Input (part 2)

Assume the plant and input are described by

$$\begin{cases} \mathbf{x}_1(t+1) & = A\mathbf{x}_1(t) + B\mathbf{u}(t) + \mathbf{v}_1(t) \\ \mathbf{y}(t) & = C\mathbf{x}(t) + \mathbf{w}_1(t) \end{cases}$$

$$\begin{cases} \mathbf{x}_2(t+1) & = F\mathbf{x}_2(t) + G\mathbf{y}(t) + \mathbf{v}_2(t) \\ \mathbf{u}(t) & = H\mathbf{x}_2(t) + J\mathbf{y}(t) + \mathbf{w}_2(t) \end{cases}$$

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This gives rise to the augmented state model to which standard KF can be applied

$$
\mathbf{x}(t+1) = \underbrace{\begin{bmatrix} A + BJC & BH \\ GC & F \end{bmatrix}}_{\mathcal{A}} \mathbf{x}(t) + \underbrace{\begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} + \begin{bmatrix} BJ & B \\ G & \end{bmatrix} \begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix}}_{\bar{\mathbf{v}}(t)}
$$

$$
\begin{bmatrix} \mathbf{y}(t) \\ \mathbf{u}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} C & \\ JC & H \end{bmatrix}}_{\mathcal{C}} \mathbf{x}(t) + \underbrace{\begin{bmatrix} I & \\ J & I \end{bmatrix}}_{\mathcal{D}} \underbrace{\begin{bmatrix} \mathbf{w}_1(t) \\ \mathbf{w}_2(t) \end{bmatrix}}_{\mathbf{w}(t)}
$$

Example with code

Before looking at more theory ... a first example of Kalman Filtering!

Jupyter notebooks:

- `1_linear_constant_velocity.ipynb`
- `2_linear_constant_velocity_with_acceleration_as_input.ipynb`
- `3_linear_constant_velocity_with_acceleration_as_state.ipynb`

The Kalman Filter: information form

1. Information variables
   - Information matrix: $Y(t|t) \triangleq P(t|t)^{-1}$
   - Information state: $\xi(t|t) \triangleq Y(t|t)\hat{\mathbf{x}}(t|t)$

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where

- $Y(t|t-1)$, $Y(t|t)$: the prediction and filtering **information** matrices
- $\xi(t|t-1)$, $\xi(t|t)$: the corresponding information states
- $\hat{\mathbf{x}}(t|t) = Y(t|t)^{-1}\xi(t|t)$

Kalman filter vs. Information filter

Standard Kalman filter

- Propagates

$$\hat{\mathbf{x}}(t|t), \quad P(t|t)$$

- Measurement update through

$$\Lambda(t) = CP(t|t-1)C^T + R$$

- Gain computation required:

$$L(t) = P(t|t-1)C^T\Lambda(t)^{-1}$$

- Natural for moderate state dimension and direct state reconstruction at each t
- Direct Covariance update

Information filter

- Propagates

$$\xi(t|t), \quad Y(t|t)$$

- Measurement update is additive:

$$\xi^+ = \xi^- + C^TR^{-1}y$$

$$Y^+ = Y^- + C^TR^{-1}C$$

- No explicit Kalman gain
- Natural for multi-sensor fusion and sparse large-scale problems
- Particularly convenient when measurements are many and independent

Key relation

$$Y(t|t) = P(t|t)^{-1}, \qquad \xi(t|t) = Y(t|t)\hat{\mathbf{x}}(t|t)$$

Why information form can be advantageous

Example: fusion of many independent sensors

Assume N sensors observe the same state:

$$y_i(t) = C_i x(t) + v_i(t), \qquad v_i(t) \sim \mathcal{N}(0, R_i), \qquad i = 1, \ldots, N$$

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- This leads to repeated gain computations and innovation covariances

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Main advantage

Updates are **simple accumulation of information**: more scalable and simpler.

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$\Longrightarrow$ relevant bound what usually referred to as **posterior (Bayesian) CRLB** (i.e., expected, wrt the posterior density, curvature of the negative posterior log-density)

$$\mathcal{I}(t|t) = -\mathbb{E}[\nabla^2_{\mathbf{x}} \log p(\mathbf{x}(t)|\mathbf{H}_t^-(\mathbf{y})) \mid \mathbf{H}_t^-(\mathbf{y})] \qquad \text{Fisher Information Matrix}$$

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- **Under the linear-Gaussian assumptions** used for KF:

$$\mathcal{I}(t|t) = Y(t|t) = P(t|t)^{-1} \quad \Longrightarrow \quad \mathrm{Cov}[\hat{\mathbf{x}}_u(t|t) - \mathbf{x}(t)] \succcurlyeq P(t|t)$$

  and the Kalman filter attains the bound:

$$\mathrm{Cov}[\tilde{\mathbf{x}}(t|t)] = \mathrm{Cov}[\mathbf{x}(t) - \hat{\mathbf{x}}(t|t)] = P(t|t).$$

Cramer-Rao limit (KF): proof of bound attainability

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&= P(t|t) + \mathrm{Cov}[d(t)] \succcurlyeq P(t|t).
\end{aligned}$$

Equality holds iff $\mathrm{Cov}[d(t)] = 0$ (i.e., $d(t) = 0$ a.s.), namely $\hat{\mathbf{x}}_u(t|t) = \hat{\mathbf{x}}(t|t)$

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Finally, we assume $S = 0$*, i.e., uncorrelated noises, to simplify the equations, i.e.* $F = A$, $\tilde{Q} = Q$, $G_t = K_t = AL_t$, $\Gamma_t = A - K_t C$.

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We recall the predictor dynamics equal to

$$
\begin{aligned}
\hat{\mathbf{x}}_{t+1} &= A\hat{\mathbf{x}}_t + K_t \left[\mathbf{y}_t - C\hat{\mathbf{x}}_t\right] \\
P_{t+1} &= A\left[P_t - P_t C^T \left(CP_t C^T + R\right)^{-1} CP_t\right] A^T + Q \\
K_t &= AP_t C^T \left[CP_t C^T + R\right]^{-1}
\end{aligned}
$$

Asymptotic Behavior: error dynamics

Also, defining the estimation error

$$\mathbf{e}_t = \mathbf{x}_t - \hat{\mathbf{x}}_t.$$

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After the update, the error dynamics are governed by

$$\begin{aligned}\mathbf{e}_{t+1} &= (A - K_t C)\mathbf{e}_t + \mathbf{v}_t - K_t \mathbf{w}_t \\ &= \Gamma_t \mathbf{e}_t + \text{noise terms}\end{aligned}$$

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Before looking at the behavior for the general case of non-stationary processes, we consider the stationary case.

Asymptotic Behavior: the stationary case

In case of stationary processes and, in particular, asymptotically stable systems, i.e., characterized by asymptotically stable state-space matrix A it is possible to state a strong direct result, connecting KF and Wiener-Kolmogorov theories as well as the asymptotic stability of the filter.

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If A is asymptotically stable then

$$\hat{\mathbf{x}}_t = \hat{\mathbb{E}}[\mathbf{x}_t | \mathbf{y}_0, \ldots, \mathbf{y}_t] \underset{t_0 \to -\infty}{\longrightarrow} \hat{\mathbf{x}}_t^{\infty}$$

where $\hat{\mathbf{x}}_t^{\infty}$ is the Wiener-Kolmogorov predictor of $\mathbf{x}_t$ based on the infinite past history of $\mathbf{y}$. Then, there exists $\lim_{t_0 \to -\infty} P_t = P_\infty$ with P_∞ solution of the Algebraic Riccati Equation (ARE). The prediction satisfies

$$\begin{aligned} \hat{\mathbf{x}}_{t+1}^{\infty} &= \Gamma_\infty \hat{\mathbf{x}}_t^{\infty} + K_\infty \mathbf{y}_t \\ K_\infty &= A P_\infty C^T \left[C P_\infty C^T + R \right]^{-1} \\ \Gamma_\infty &= A - K_\infty C \end{aligned}$$

And if $(\Gamma_\infty, K_\infty)$ is reachable, then Γ_∞ is asymptotically stable.

Asymptotic Behavior: structural assumptions

In the more general case of non-stationary processes (A not asymptotically stable), the behavior is studied through the behavior of P_t (i.e. of the RE)

Yet, a couple of assumptions are needed...

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These conditions (checked via PBH tests[7]) are natural because:

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$R \succ 0$, detect. of (A, C), stab. of $(A, Q^{1/2}) \implies \exists!$ stabilizing steady-state P_∞

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Asymptotic Behavior: fundamental theorem of KF theory

Necessary and sufficient condition for:

- $\exists!$ a symmetric P_∞ solution of the A.R.E.

$$P = APA^T - APC^T(CPC^T + R)^{-1}CPA^T + Q$$

- the solution P_∞ is stabilizing
- $\lim_{t\to\infty} P_t = P_\infty \quad \forall$ initial symmetric positive semidefinite P_0

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Interpretation:

- after transients, the filter "forgets" the initial covariance,
- the gain becomes constant,
- the estimation error reaches a stationary covariance regime.

Asymptotic Behavior: steady-state KF

Analogously to the stationary case, once P_k and K_k have converged, one can replace the time-varying filter by the constant-gain filter

$$\begin{aligned} \hat{\mathbf{x}}_{t+1|t} &= A\hat{\mathbf{x}}_{t|t}, \\ \hat{\mathbf{x}}_{t+1|t+1} &= \hat{\mathbf{x}}_{t+1|t} + K_\infty(\mathbf{y}_{t+1} - C\hat{\mathbf{x}}_{t+1|t}). \end{aligned}$$

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Advantages of the constant-regime filter:

- lower online computational cost,
- simpler implementation in embedded systems,
- explicit LTI error dynamics,
- same asymptotic performance as the full Kalman filter.

Hence, for time-invariant models, the practical end-point is often a *steady-state Kalman filter*.

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The tuning parameters shape this regime:

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The asymptotic gain is the precise mathematical embodiment of the model/measurement tradeoff.

Asymptotic Behavior: continuous-time counterpart

For the continuous-time model

$$\dot{x} = Ax + Gw, \qquad y = Cx + v,$$

with white noises of intensities Q and R, the covariance satisfies the differential Riccati equation

$$\dot{P} = AP + PA^\top + GQG^\top - PC^\top R^{-1} CP.$$

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The constant-regime filter is then the steady-state Kalman–Bucy filter.

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4. The estimation error covariance converges: $P_k \to P_\infty$
5. The asymptotic filter becomes a constant-gain linear filter with the same steady-state performance as the full time-varying Kalman filter.

`4_asymptotic_behavior.ipynb`

3 Nonlinear Theory

Linear vs Nonlinear Kalman Filtering

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  - **Unscented Kalman Filter (UKF)**: Uses deterministic sampling to capture mean and covariance of nonlinear transformations.
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Estimator	**Model**	**Assumed Distribution**	**Computational Cost**
KF	Linear	Gaussian	Low
EKF	Locally Linear	Gaussian	Low/Medium (depends on Jacobians)
UKF	Nonlinear	Gaussian	Medium
PFs	Nonlinear	Arbitrary	High

Extended Kalman Filter (EKF): essentials and notation

- Linearizes nonlinear functions $f(x, u)$ and $h(x)$ around current estimate.
- Same equations as linear KF, but with linearized matrices.
- Limitations: Linearization errors, Jacobian computation.

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Consider a discrete-time[8] non-linear state model of the form

$$\begin{aligned} \mathbf{x}(k+1) &= f(\mathbf{x}(k), \mathbf{u}(k)) + \mathbf{v}(k), \quad \mathbf{v} \sim \mathcal{N}(0, Q) \\ \mathbf{y}(k) &= h(\mathbf{x}(k)) + \mathbf{w}(k), \quad \mathbf{w} \sim \mathcal{N}(0, R) \end{aligned}$$

under the usual assumptions $\mathbf{v} \perp \mathbf{w} \perp \mathbf{x}_0$,

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under the usual assumptions $\mathbf{v} \perp \mathbf{w} \perp \mathbf{x}_0$, and denote the Jacobian matrices

$$F_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k|k}} \qquad H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{k+1|k}}$$

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EKF: discrete-time equations

1. Prior estimates
   - State: $\hat{\mathbf{x}}(k+1|k) = f(\hat{\mathbf{x}}(k|k), \mathbf{u}(k))$
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Nonlinearity used to properly propagate the prior estimate and the measurement model to compute the pseudo innovation.

EKF: generalization - continuous-discrete equations

The EKF is also used in more general setups of continuous-discrete systems.

[9] Matrix satisfying the Volterra equation which, for constant F, equals $\Phi(t,s) = e^{F(t-s)}$.

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Consider a continuous-time state-model with sampled measurements at discrete instants t_k (possibly non uniform)

$$\begin{aligned}\dot{\mathbf{x}}(t) &= f(\mathbf{x}(t)) + \xi(t) \\ \mathbf{y}(t_k) &= h(\mathbf{x}(t_k)) + \mathbf{w}(t_k)\end{aligned}$$

being ξ white noise of intensity $Q \geq 0$; $\mathbf{w}(t_k)$ white of variance $R > 0$ and $\{\xi(t)\} \perp \{\mathbf{w}(t_k)\} \perp \mathbf{x}(t_0)$. $f, h \in \mathbb{C}^1$.

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Let $\Phi(t, s, \overline{\mathbf{x}})$ be the fundamental matrix[9] associated to the time-varying linear system $\dot{\mathbf{z}} = F(\overline{\mathbf{x}}(t))\mathbf{z}$ where $F(\overline{\mathbf{x}}(t)) = \frac{\partial f}{\partial x}|_{\mathbf{x}=\overline{\mathbf{x}}(t)}$ and define

$$Q(t_k) := \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, s, \overline{\mathbf{x}}) Q \Phi(t_{k+1,s}, \overline{\mathbf{x}})^T ds$$

Under these conditions, the equations generalize as follows.

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  - Set $\hat{\mathbf{x}}(k+1|k) = \overline{\mathbf{x}}(t_{k+1})$
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  - Alternative CT Riccati: $\dot{P}(t) = F(t)P(t) + P(t)F(t)^T + Q(t)$

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     - $\Lambda(k+1) = H(k+1|k)P(k+1|k)H(k+1|k)^T + R$
     - $H(k+1|k) := H(\hat{\mathbf{x}}(k+1|k)) = \frac{\partial h}{\partial x}|_{\mathbf{x}=\hat{\mathbf{x}}(k+1|k)}$
   - Variance: $P(k+1|k+1) = (I - L(k+1)H(k+1|k))P(k+1|k)$

with initial conditions $\hat{\mathbf{x}}(0,-1) = \mathbb{E}[\mathbf{x}(t_0)]$, $P(0|-1) = \mathrm{Var}[\mathbf{x}(t_0)]$

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To avoid explicit integration, when the signals are sufficiently slow w.r.t. the sampling time, a full DT scheme is used, by discretizing the CT dynamics, e.g.,

$$\mathbf{x}(k+1) = \hat{f}_k(\mathbf{x}(k)) \stackrel{Euler}{=} \mathbf{x}(k) + (t_{k+1} - t_k) f(\mathbf{x}(k))$$

also used to propagate the prior estimated state in the filter equations

$$\hat{\mathbf{x}}(k+1|k) = \hat{f}_k(\hat{\mathbf{x}}(k|k))$$

All other equations remain the same assuming now $\Phi(k|k) = \frac{\partial \hat{f}_k}{\partial x}|_{\mathbf{x}=\hat{\mathbf{x}}(k|k)}$.

`5_nonlinear_ekf.ipynb`

Unscented Kalman Filter (UKF)

- Uses sigma points to capture higher-order moments[10]: for any random vector $\mathbf{x}$, sigma points are a set of any N vectors satisfying:
  - first order: weighted average of points corresponds to $\mathbb{E}[\mathbf{x}]$
  - second order: weighted average of points covariance corresponds to $\mathrm{Var}[\mathbf{x}]$

  The critical point is to find the right choice of weights: for this standard approaches exists.

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  - better handling of nonlinearities since P is estimated considering nonlinear transforms.

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UKF equations are not difficult but heavy in notation. We'll see them in code rather than on slides. `9_nonlinear_ukf.ipynb`

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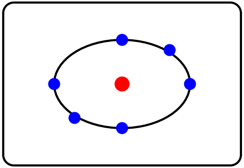
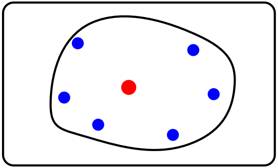
UKF: graphical representation

Prior state distribution



propagate sigma points $\xrightarrow{\hspace{2em}}$ through $f(\cdot)$

Predicted sigma points

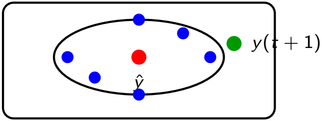


$\hat{\mathbf{x}}(t|t)$, $P(t|t)$ and sigma points

Estimate $\hat{\mathbf{x}}(t+1|t)$, $P(t+1|t)$ as combinations of transformed sigma points and compute new sigma points.

propagate points through $h(\cdot)$

Update $\hat{\mathbf{x}}(t+1|t+1)$, $P(t+1|t+1)$ with $\mathbf{y}(t+1)$



Predict measurement as weighted combination of transformed points

Particle Filters (PFs)

Particle Filters are a family rather than a single estimation algorithm.

General idea:

- use many samples/particle
- use random sampling to approximate any underlying distribution
- often exploit a *genetic* approach consisting of a selection/updating step alternate with a mutation/prediction step

Common implementations are:

- Markov Chain Monte Carlo
- Sequential Importance Sampling (SIS)
- Sequential Importance Resampling (SIR)

3. Adds-on

A Practical Recipe for Tuning a Kalman Filter (part 1)

Goal: essentially, tune the two covariances that matter most:

$$Q \quad \text{(process noise)} \qquad R \quad \text{(measurement noise)}$$

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1. Estimate measurement noise R from real data

- Hold the system still or run in a regime with known truth
- Compute sample covariance of sensor error

$$R \approx \mathrm{cov}(y_k - y_k^{\mathrm{true}}) \quad \text{or approximately} \quad R \approx \mathrm{cov}(y_k - \bar{y})$$

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Start with simple diagonal Q and then move towards $Q = \alpha Q_0$, where Q_0 encodes relative trust between states and α is one scalar to tune.

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3. Check the innovations

$$\nu_k = y_k - H\hat{x}_k^-, \qquad S_k = HP_k^- H^\top + R$$

A good filter has innovations that are approximately white, zero-mean, and with the expected size (covariance close to S_k).

A Practical Recipe for Tuning a Kalman Filter (part 2)

4. Tune using observed behavior

Symptom	**Adjustment**
Estimate too noisy	increase R or decrease Q
Estimate too sluggish	increase Q or decrease R
Bias / drift not tracked	increase Q on bias states

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Estimate too noisy	increase R or decrease Q
Estimate too sluggish	increase Q or decrease R
Bias / drift not tracked	increase Q on bias states

Best practical rule: fix R from measurements, then tune only Q until the filter is as fast as possible without visibly fitting noise.

Example sweep

```
for alpha in [1e-3, 1e-2, ..., 1e3]:
    Q = alpha * Q0
    run filter
    inspect innovation variance + tracking lag
```

`8_filter_tuning.ipynb`

Fixed-Interval Rauch-Tung-Striebel Smoother

Objective of smoothing:

- Fixed-lag: provide an almost real-time (slightly delayed) improved estimate at $k - N$ using a sliding window of measurements up to k.
- Fixed-interval: provide the best estimate using all available measurements up to a certain instant n.

[11] There are also other smoother implementation e.g., the Modified Bryson-Frazier, which we are not going to cover.

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The RTS smoother[11] consists of:

- forward pass: exactly equal to the standard KF equations to compute $\hat{\mathbf{x}}(k|k-1)$, $\hat{\mathbf{x}}(k|)$, $P(k|k-1)$ and $P(k|k)$.

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- backward pass: compute the *smoothed* estimates $\hat{\mathbf{x}}(k|n)$, $P(k|n)$ via
  - $\hat{\mathbf{x}}(k|n) = \hat{\mathbf{x}}(k|k) + C_k\left(\hat{\mathbf{x}}(k+1|n) - \hat{\mathbf{x}}(k+1|k)\right)$
  - $P(k|n) = P(k|k) + C_k\left(P(k+1|n) - P(k+1|k)\right)C_k^T$
  - $C_k = P(k|k)F^T P(k+1|k)^{-1}$

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Note that, conversely to fixed-lag smoothers, fixed-interval are **offline** algorithms as they require all future measurements up to n.

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LQG control and its connection to KF

Problem setup (stochastic optimal control)

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad y_t = Cx_t + v_t, \quad w_t \sim \mathcal{N}(0, Q), \quad v_t \sim \mathcal{N}(0, R)$$

Objective (LQG cost): $J = \mathbb{E}\left[\sum_{t=0}^{T} x_t^T Q_x x_t + u_t^T R_u u_t\right]$

Key result (Separation principle): $\boxed{\text{Optimal control} = \text{LQR} + \text{KF}}$

- **State estimation (Kalman filter)**: $\hat{\mathbf{x}}(t|t) = \mathbb{E}[\mathbf{x}_t \mid \mathbf{H}_t^-(\mathbf{y})]$
- **Control law (LQR)**: $u_t = -K_t\hat{\mathbf{x}}(t|t)$ (Controller uses $\hat{\mathbf{x}}$, not $\mathbf{x}$)

Dual Riccati equations (forward + backward)

- KF covariance: $P_{t+1} = AP_tA^T + Q - AP_tC^T(CP_tC^T + R)^{-1}CP_tA^T$
- LQR cost2go: $S_t = A^TS_{t+1}A + Q_x - A^TS_{t+1}B(B^TS_{t+1}B + R_u)^{-1}B^TS_{t+1}A$

KF (estimation)	LQR (control)
$A,\ C$	$A^T,\ B^T$
$Q,\ R$	$Q_x,\ R_u$
P_t	S_t
K_t^{KF}	K_t^{LQR}

Interpretation

- KF: optimal *information extraction*
- LQR: optimal *control action*
- LQG: optimal closed-loop system

Additional resources

- https://github.com/mintisan/awesome-kalman-filter?tab=readme-ov-file
- https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python
- https://kalmanfilter.net/

End! Thank you!